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Specializing in Information Technology

AI-Powered Civil Law Assistance Platform for Sri Lankan Citizens

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Introduction

- Sri Lankan citizens have been struggling to have access to legal information.
- People often seek advice from lawyers or police officers.
- Sometimes incorrect advice can lead bad decisions.
- The civil law domains like property, vehicles & highways, and marriage & divorce require proper guidance. Currently available systems do not provide ease of use.
- A legal assistant service for understanding questions in Sinhala /English and offering basic trustworthy legal information is also brought by this project.

Importance & Impact of the Solution

- Improves access to legal information for Sri Lankan citizens
- Helps people understand laws in **simple language**
- Reduces dependence on incorrect or informal legal advice
- Supports **Sinhala and English** users
- Focuses on common civil law areas:
  - Property
  - Vehicles & highways
  - Marriage & divorce
- Saves time and effort when seeking basic legal guidance
- Increases legal awareness among the general public

Existing Systems

LankaLaw

- Online legal database
- Mainly used by lawyers and law students
- Not easy for normal citizens

• LawMate

- AI-based legal platform
- Mainly supports lawyers and legal professionals
- Limited support for general public users

RESEARCH GAP

- Designed **specifically for ordinary Sri Lankan citizens**
- Explains legal information in **simple Sinhala and English**
- Uses an **AI-powered conversational chatbot**
- Focuses only on **common civil law areas**:
  - Property
  - Vehicles & highways
  - Marriage & divorce
- Provides **trustworthy and verified legal information**
- Includes a **smart lawyer recommendation feature**

RESEARCH PROBLEM

Ordinary Sri Lankan citizens struggle to access and understand legal information due to complex legal language and professional-focused platforms. Existing digital legal systems are designed mainly for lawyers and students and do not provide simple, multilingual explanations or conversational support. As a result, people lack legal awareness, depend on unreliable advice, and face difficulties in finding suitable lawyers, especially in rural areas. This highlights the need for an accessible, AI-powered legal assistance platform tailored for the general public.

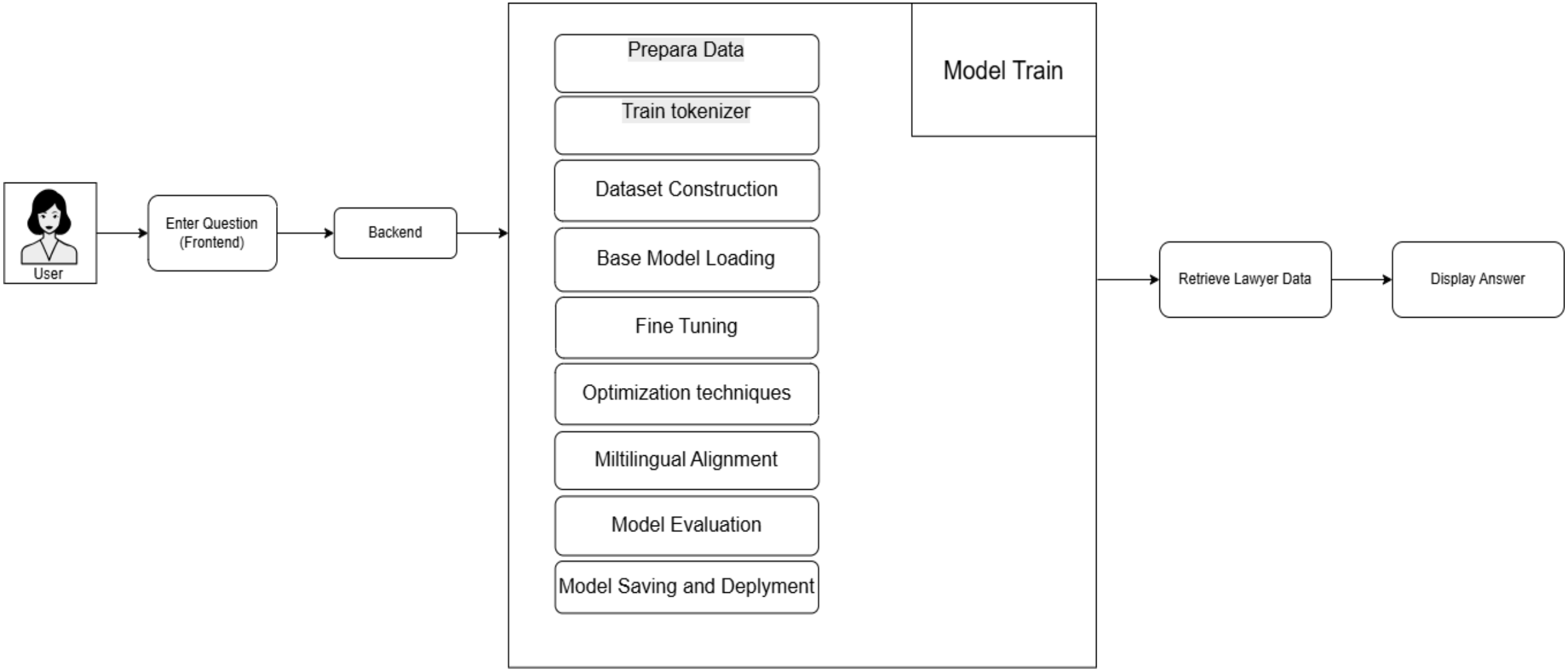
Main Objective

Develop an **AI-powered, multilingual legal assistance platform** that provides **simple legal explanations.**

Sub Objectives

- Implement a legal information retrieval system using Sri Lankan legal sources
- Build a lawyer recommendation
- Design a user-friendly web application
- Evaluate the system to **improve legal awareness**, focusing on accuracy, usability, and impact

Overall System Diagram



```
trainer.train()
```

```
The model is already on multiple devices. Skipping the move to device specified in `args`.
==((====))==  Unsloth - 2x faster free finetuning | Num GPUs used = 1
   \\   /|    Num examples = 2 | Num Epochs = 1,000 | Total steps = 1,000
O^O/ \_/ \    Batch size per device = 1 | Gradient accumulation steps = 4
\        /    Data Parallel GPUs = 1 | Total batch size (1 x 4 x 1) = 4
 "-____-"     Trainable parameters = 13,631,488 of 8,043,892,736 (0.17% trained)
Unsloth: Will smartly offload gradients to save VRAM!
[ 525/1000 1:12:53 < 1:06:12, 0.12 it/s, Epoch 524/1000]
```

Step	Training Loss
10	

Proof of Work

legal_chatbot_db | lawyers | qa_chunks | +

Cluster0 > legal_chatbot_db

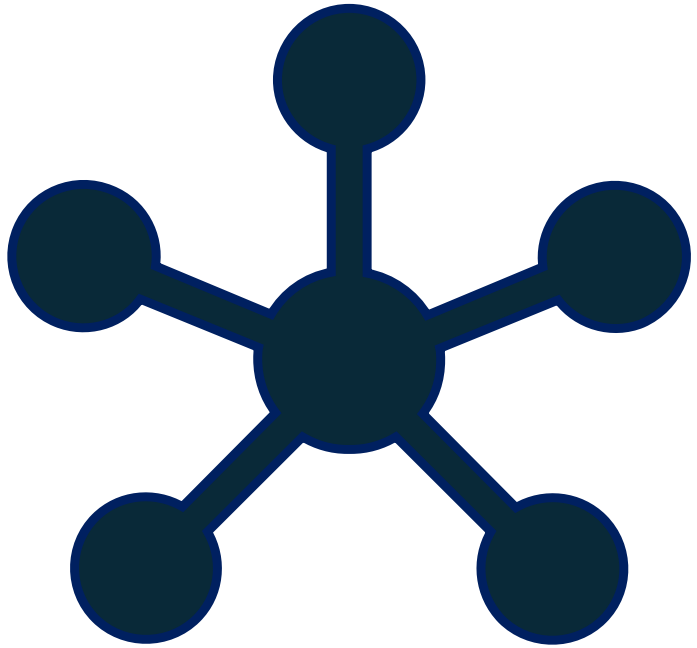
View monitoring | Visualize your data | + Create collection | Ref

Collection name	Properties	Storage size	Document	Avg. document size	Indexes	Total index size
lawyers	-	36.86 kB	2	193.00 B	1	36.86 kB
qa_chunks		45.06 kB	3	9.90 kB	2	73.73 kB

Proof of Work

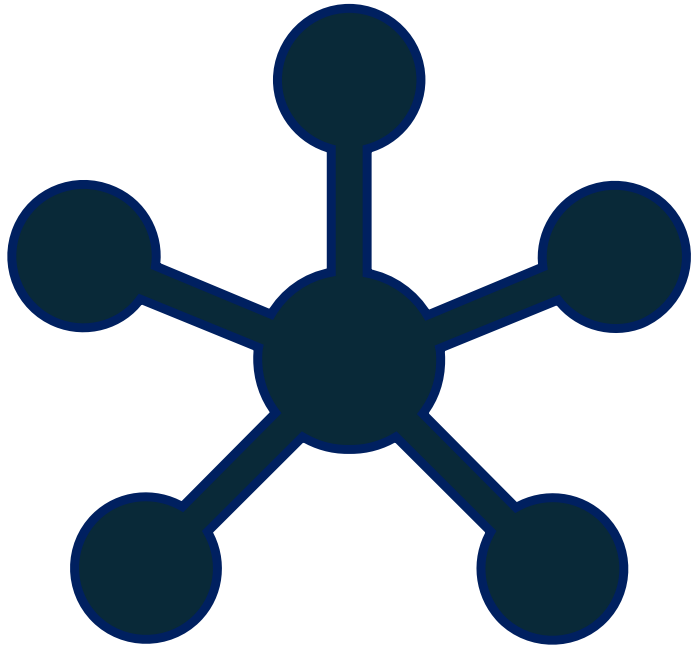
- Property Laws
- Vehicle and Highway Laws
- Marriage and Divorce Laws
- Lawyers

Application of key pillars in the specialized area of knowledge



- **Artificial Intelligence (AI):**
  - Used to analyze user queries and generate legal explanations
- **Natural Language Processing (NLP):**
  - Enables Sinhala–English language understanding and interaction
- **Data Management & Information Systems:**
  - Manages and retrieves structured Sri Lankan legal data
- **Human Computer Interaction (HCI):**
  - Ensures usability and accessibility for non-technical users
- **Software Engineering:**
  - Applies systematic design, development, and testing practices

Application of technologies in the relevant key pillar



- **Large Language Models (LLMs):**
  - Used to implement the Sinhala–English legal chatbot
- **Machine Learning Techniques:**
  - Improve response relevance and accuracy
- **Web Technologies (React, Node.js):**
  - Used to develop the front-end and back-end of the system
- **Databases & Data Modeling:**
  - Store legal datasets, user queries, and lawyer information

FUNCTIONAL REQUIREMENTS

- Users can ask legal questions in **Sinhala or English**
- System provides **simple legal explanations** using AI chatbot
- Support legal domains:
  - Property
  - Vehicles & highways
  - Marriage & divorce
- Recommend suitable lawyers based on question.
- Allow users to access the system through a **web application.**

NON-FUNCTIONAL REQUIREMENTS

- **Usability**
- **Performance**
- **Accuracy**
- **Security**
- **Availability**
- **Maintainability**

Completion of the project

A fully functional AI-powered civil law assistant was developed using **only custom-trained models**.

The system provides fast, understandable, and citation-based legal information.

Demonstrates feasibility of building a legal NLP model without pretrained systems.

Supports Sinhala/English and covers three key domains: property, vehicle & highway, and marriage & divorce.

Positioned as a scalable foundation for future legal tech solutions in Sri Lanka.

- Risk: Inaccurate or misleading legal responses
  - Mitigation: Use verified Sri Lankan legal sources and validation checks
- Risk: Sinhala–English language understanding errors
  - Mitigation: Train and test the LLM using bilingual legal datasets
- Risk: Limited or incomplete legal datasets
  - Mitigation: Integrate multiple trusted sources and update regularly
- Risk: User trust and system misuse
  - Mitigation: Provide disclaimers and restrict system to informational guidance
- Risk: System performance and scalability issues
  - Mitigation: Optimize backend and plan for scalable deployment
- Risk: Difficulty in user adoption
  - Mitigation: Apply HCI principles and conduct usability testing

References

- [1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online].
  Available: https://proceedings.neurips.cc/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf

- [2] U. Kanewala, S. Weerakoon and R. Nawarathna, "Exploring Deep Learning and Tree-Based Ensemble Models for Chiller Energy Consumption Predictions," 2021 IEEE 16th International Conference on Industrial and Information Systems (ICIIS), Kandy, Sri Lanka, 2021, pp. 306-311, doi: 10.1109/ICIIS53135.2021.9660662. keywords: {Radio frequency;Deep learning;Energy consumption;Recurrent neural networks;Buildings;Predictive models;Energy efficiency;HVAC Systems;Energy Consumption;Machine Learning;Deep Learning;Feedforward Neural Networks;Random Forrest;Recurrent Neural Networks},

COMMERCILATION

Customer Persona

Empowering Sri Lankan citizens with easy legal guidance while helping lawyers connect with the right clients.

Market Size

Serving a large and growing population facing everyday legal challenges across Sri Lanka.

What is unique in your Solution?

A smart Sinhala–English AI legal chatbot with built-in lawyer matching, designed for local laws and people.

How will the cost be recovered?

Free access for the public, with premium features and subscription plans for legal professionals.

THANK YOU !